

Different Notions of Capacity for Quantum Channels

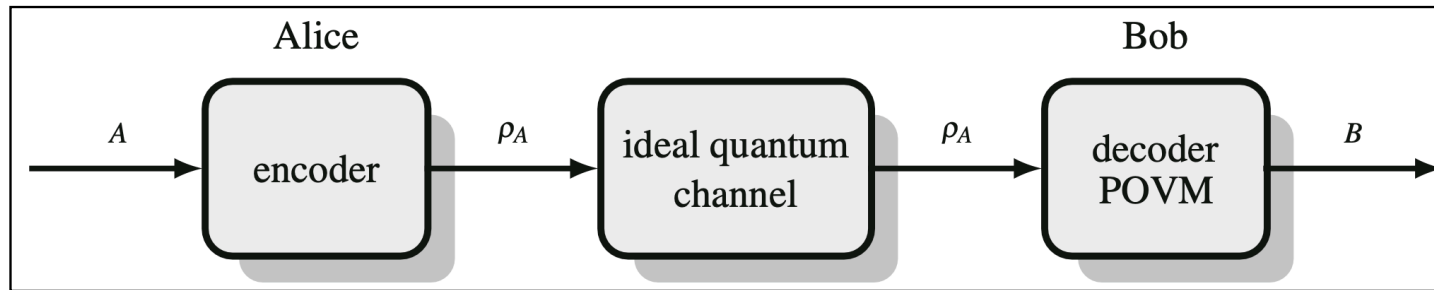


- **Classical Capacity $C(\mathcal{E})$:** Best rate at which classical information (bits) can be transmitted over the channel.
- **Entanglement-Assisted Classical Capacity $C_E(\mathcal{E})$:** Best rate at which classical information (bits) can be transmitted over the channel when the sender and receiver share an arbitrary number of quantum states (ebits).
- **Private Classical Capacity $P(\mathcal{E})$:** Best rate for sending classical information to achieve high fidelity between sender and receiver without leaking information to the environment (Eve). Applicable for secret key capacity of QKD and quantum wiretap channel capacity.
- **Quantum Capacity $Q(\mathcal{E})$:** Best rate for sending quantum information (qubits) over the channel.



Classical Capacity: Accessible Information

- **Classical-Quantum Communications:** Classical information transfer Alice $\rightarrow$ Bob with classical source at Tx.: $A \simeq (\mathcal{A}, p_A)$ and receiver side sees a classical source: $B \simeq (\mathcal{B}, p_{A|B})$



- **Mutual Information:** $I(A; B) = H(A) - H(A | B)$
quantifies how much information A and B have in common
- **Accessible Information:** The maximum mutual information obtained through optimum POVM measurement

$$I_{\text{acc}} := \max_{\text{POVM}} I(A; B)$$



Holevo Information

- **Holevo Entropy:** Ensemble of K quantum states $\mathcal{L} = \{(p_1, \rho_1), \dots, (p_K, \rho_K)\} = \{p_a, \rho_a\}$ then χ -information or Holevo- χ is defined as:

$$\chi(\{p_a, \rho_a\}) := S(\rho) - \sum_a p_a S(\rho_a)$$

$$\text{where } \rho = E[\rho_A] = \sum_a p_a \rho_a \text{ (Avg. Density operator)}$$

- **Holevo- χ properties:**

1. **Entropy of the constellation:** Holevo- χ depends only on the ensemble $\mathcal{L}$ i.e., $\chi(\mathcal{L})$
2. **Non-negativity:** $\chi(\mathcal{L}) \geq 0$ (due to concavity of quantum entropy).
3. **For pure states:** $S(\rho_a) = 0$, thus $\chi(\{p_a, \rho_a\}) = S(\rho)$
4. A quantum operation Φ can never increase the Holevo- χ , i.e., $\mathcal{L}' = \{(p_a, \Phi(\rho_a))\}$ then $\chi(\mathcal{L}') \leq \chi(\mathcal{L})$



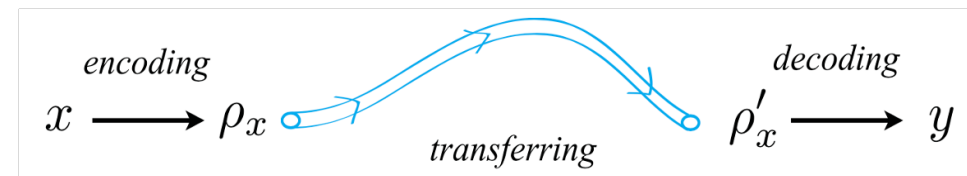
Holevo Bound

- **Theorem (Holevo's bound):** For any measurement Bob can do, the mutual information is limited by the Holevo- χ :

$$I(A; B) \leq \chi(\{p_a, \rho_a\})$$

- The conditions for the equality in Holevo Bound are:
 1. The density operators ρ_a commute (are simultaneously diagonalizable) and
 2. Bob perform the **POVM** measurement with the common eigenbasis of the $\{\rho_a\}$
- **Classical Quantum Communication:**
- **HSW Theorem:** Classical capacity of quantum channels

$$C(\mathcal{E}) = \max_{p_x, \rho_x} \left\{ S \left(\sum_x p_x \mathcal{E}[\rho_x] \right) - \sum_x p_x S(\mathcal{E}[\rho_x]) \right\}$$



Quantum Channel : $\mathcal{E}[\cdot]$

$$\rho'_x = \mathcal{E}[\rho_x]$$



Example: Holevo Bound

- **Example:** A qubit system where Alice prepares $|\psi_0\rangle = |0\rangle$, $|\psi_1\rangle = \cos \theta |0\rangle + \sin \theta |1\rangle$ with prob. $p_0 = p$ and $p_1 = 1 - p$.

- Density operators: $\rho_0 = |\psi_0\rangle \langle \psi_0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\rho_1 = |\psi_1\rangle \langle \psi_1| = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$

- Avg. density operator: $\rho = p\rho_0 + (1 - p)\rho_1 = \begin{bmatrix} (1 - p)\cos^2 \theta + p & (1 - p)\cos \theta \sin \theta \\ (1 - p)\cos \theta \sin \theta & (1 - p)\sin^2 \theta \end{bmatrix}$

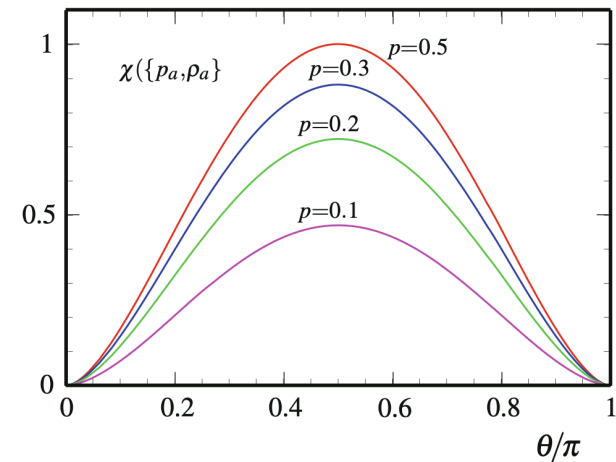
- Note for pure states: $S(\rho_0) = S(\rho_1) = 0$

- Eigenvalues of ρ :

$$\lambda_{\pm} = \frac{1}{2} \left[1 \pm \sqrt{2p^2 - 2p(p - 1)\cos 2\theta - 2p + 1} \right]$$

then:

$$\chi(p_a, \rho_a) = S(\rho) = H(\{\lambda_-, \lambda_+\})$$



Holevo bound as a function of the parameter θ in a qubit system

Maximum is obtained when the states are orthogonal, i.e., for $\theta = \pi/2$



Holevo Capacity Examples

- Ex-1: Identity (Noiseless) Qubit Channel: $\mathcal{N}(\rho) = \rho$
- Ex-2: Completely Depolarising Qubit Channel: $\mathcal{N}(\rho) = \frac{I}{2}$
- Ex-3: Bit Flip Channel: $\mathcal{N}(\rho) = (1 - p)\rho + pX\rho X$
- Ex-4: Depolarising Qubit Channel: $\mathcal{N}(\rho) = (1 - p)\rho + p\frac{I}{2}$



- Ex-1: Identity (Noiseless)
Qubit Channel: $\mathcal{N}(\rho) = \rho$

Holevo Capacity of Q. Channels.

(i) Noiseless Channel $\boxed{\mathcal{N}(\rho) = \rho}$ $\xrightarrow{\rho} \boxed{\mathcal{N}} \xrightarrow{\rho'}$

Choose orthonormal input states. $\frac{|0\rangle}{\sqrt{2}}, \frac{|1\rangle}{\sqrt{2}}$

$$\rho_0 = |0\rangle\langle 0|, \rho_1 = |1\rangle\langle 1|$$

$$\rho'_0 = \mathcal{N}(\rho_0) = |0\rangle\langle 0| \quad \& \quad \rho'_1 = \mathcal{N}(\rho_1) = |1\rangle\langle 1|$$

Capacity $\rightarrow$ Find Holevo χ of $\left\{ \frac{\rho'_0}{\sqrt{2}}, \frac{\rho'_1}{\sqrt{2}} \right\}$

$$S(\rho'_0) = 0 \quad S(\rho'_1) = 0$$

$$\begin{aligned} \rho &= \frac{1}{2} \rho'_0 + \frac{1}{2} \rho'_1 = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1| = \mathbb{I}/2 \\ &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \end{aligned}$$

$$S(\rho) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = 1 \quad \leftarrow$$

$$\text{Capacity} = S(\rho) - \sum p_a S(\rho'_a) \rightarrow 0$$

$$\rightarrow \boxed{= 1 \text{ bit/qubit ch. use}} \rightarrow \underline{\text{Holevo's Bound.}}$$

$\&$ By using ^upre-shared entanglement + ^u1 use of Noiseless Qubit Channel

$\hookrightarrow$ Superdense Coding. $\equiv 2 \text{ bit/qubit ch. use Capacity} \checkmark$



- Ex-2: Completely Depolarising Qubit Channel:

$$\mathcal{N}(\rho) = \frac{I}{2}$$

Ex-2: Completely depolarizing channel $\boxed{\mathcal{N}(\rho) = I/2}$
 Alice sends $\left\{ \underset{\gamma_2}{|0\rangle}, \underset{\gamma_2}{|1\rangle} \right\}$

$$\rho_0 = |0\rangle\langle 0| \quad \rho_1 = |1\rangle\langle 1|$$

$$\left. \begin{array}{l} \rho_0 \longrightarrow \rho_0' = I/2 \\ \rho_1 \longrightarrow \rho_1' = I/2 \end{array} \right\} \text{Holds no information}$$

$$S(\rho_0') = 1, \quad S(\rho_1') = 1.$$

$$\rho = \frac{1}{2} \rho_0' + \frac{1}{2} \rho_1' = \frac{1}{2} \frac{I}{2} + \frac{1}{2} \frac{I}{2} = \frac{I}{2}$$

$$S(\rho) = 1$$

$$\begin{aligned} \chi &= S(\rho) - \sum p_i S(\rho_i) \\ &= 1 - \left(\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 \right) = 0 \end{aligned}$$

$C(\mathcal{N}) = 0 \rightarrow$ Channel is destroying all the information.

$\rightarrow$ No distinguishability survives